Pratyush Aman

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Education

- **Indian Institute of Science Education and Research, Thiruvananthapuram** 2021–2026

BS-MS Dual Degree in Interdisciplinary Chemistry

- **Relevant Coursework:** Quantum Chemistry, Chemical Kinetics, Computational Chemistry, Computational Chemical Biology, Coordination Chemistry

- **St. Karen's High School, Patna (CBSE)** 2018–2020

Class XII, Science Stream — 86.4%

- **Subjects:** Physics, Chemistry, Mathematics, English, Informatics Practices

Research Experience

- **MS Major Project: Burst Oscillations in a Periodically Forced Supramolecular System** Aug 2025–Jul 2026

IISER Thiruvananthapuram | Supervisor: Dr. Pushpita Ghosh

- Built a four-variable kinetic model of fuel-driven supramolecular polymerization and reduced it exactly to a three-variable system using a conservation law, using linear stability and eigenvalue analysis to locate its Hopf bifurcation regimes.
- Showed that periodic parametric forcing of the model produces *burst oscillations*, with alternating active and silent phases as the trajectory repeatedly crosses a Hopf boundary; applied slow–fast analysis to link burst onset to this slow passage, giving a burst threshold set by material (rate-constant) parameters rather than by the forcing.
- Constructed Arnold tongue diagrams over forcing amplitude–frequency grids using Poincaré stroboscopic sections, mapping frequency-locking regimes and phase-locking (devil's staircase) structure.
- Derived an analytical Hopf bifurcation surface for the reduced model in Mathematica, and extended the model to spatiotemporal dynamics with combined periodic forcing and Gaussian stochastic perturbations, observing pattern formation during the assembly–disassembly process alongside the forced oscillatory dynamics.

- **DFT Analysis Internship** May–Jul 2024

IIT Hyderabad | Supervisor: Dr. Debasis Koner

- Performed geometry optimization and electronic structure analysis of H_2O, H_3O^+ and transition metal complexes using DFT in Gaussian.

- **Molecular Simulation Internship** Winter 2024

IIT Madras | Supervisor: Dr. Raghunathan Rengaswamy

- Implemented molecular energy calculations in Python and analyzed Lennard–Jones potential models.

- **Protein & Membrane Analysis Coursework**

IISER Thiruvananthapuram

- Ran MD simulations of a GB1 hairpin, a polymer, and membrane systems; built solvated systems in CHARMM-GUI and performed molecular docking.
- Predicted protein structures with AlphaFold, I-TASSER and RoseTTAFold.

Manuscripts

- **Pratyush Aman and Pushpita Ghosh**, "Burst Oscillations in a Periodically Forced Chemically Fueled Supramolecular System," *manuscript in preparation (to be submitted October 2026).*

Independent Research Projects

- **Scientific Machine Learning for Nonlinear Dynamics**

Self-initiated

- Developed and evaluated a physics-informed SciML framework using FiLM neural networks and physical constraints across the Van der Pol, FitzHugh–Nagumo, Lorenz, and supramolecular oscillator systems.
- Demonstrated improved bifurcation, long-term dynamical, conservation, and out-of-distribution fidelity compared with unconstrained neural models.
- Applied Jacobian/eigenvalue analysis, Lyapunov exponents, invariant measures, uncertainty quantification, and capacity/ablation studies.
- Code and reproducibility: [GitHub]

- **GNN-based Molecular Property Prediction**

Self-initiated

- Developed graph neural networks in PyTorch Geometric and RDKit for solubility and logP prediction of small molecules.
- Implemented atom-level attention to make predictions chemically interpretable.

Technical Skills

Dynamics & Modeling:	Bifurcation analysis, slow–fast methods, Arnold tongues, ODE/PDE simulation
Scientific Computing:	Python (NumPy, SciPy, Pandas, Matplotlib), Mathematica, C++, Bash Scripting(automation)
Computational Chemistry:	Gaussian, ORCA, GROMACS, VMD, CHARMM-GUI
Machine Learning:	PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, RDKit
Software & Tools:	Git, Jupyter, Overleaf, OriginLab, Linux/Unix

Conferences & Presentations

- **COLD Winter School 2026 (Complexity, Optimisation, Learning, and Dynamics)**, Centre for Complex Systems and Dynamics, IIT Madras — selected through a competitive process among 30 students.
- **International Conference on Nonlinear Systems and Dynamics (CNSD 2026)**, IISER Thiruvananthapuram — participant.
- **Anvesha 2023**, IISER Thiruvananthapuram — *AstroChemistry: XPS and Chemical Curvature.*
- **Anvesha 2022**, IISER Thiruvananthapuram — *Calculation of Lagrange Points using Space-Time Curvature.*

Certifications

- **Claude Code in Action**, Anthropic, Sep 2026. [Certificate]
- **Claude with the Anthropic API**, Anthropic, Sep 2026. [Certificate]

References

Dr. Pushpita Ghosh
MS Thesis Supervisor
Assistant Professor, School of Chemistry
IISER Thiruvananthapuram
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Dr. Jagannath Mondal
Instructor, Computational Chemical Biology
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